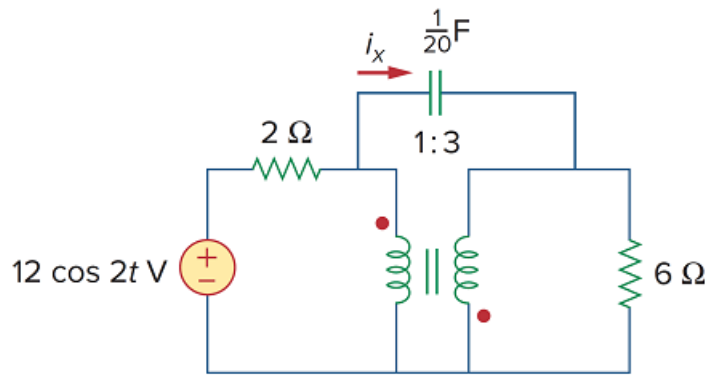


Problem

Find current i_x in the ideal transformer circuit shown in Fig. 13.114.

Figure 13.114



Step-by-step solution

Step 1 of 8

Refer to figure 13.114 given in the textbook.

Redraw the circuit the circuit as shown in figure 1.

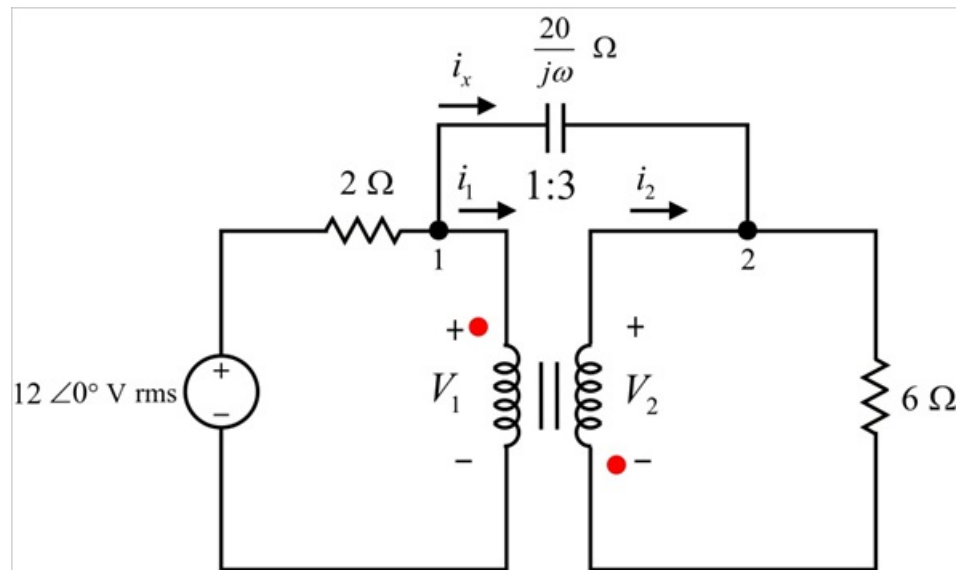


Figure 1

Step 2 of 8

Apply Kirchhoff's current law at node 1 of the circuit shown in figure 1.

$$\frac{V_1 - 12}{2} + \frac{V_1 - V_2}{\left(\frac{20}{j\omega}\right)} + i_1 = 0 \quad \dots\dots (1)$$

Apply Kirchhoff's current law at node 2 of the circuit shown in figure 1.

$$-i_2 + \frac{V_2 - V_1}{\left(\frac{20}{j\omega}\right)} + \frac{V_2}{6} = 0 \quad \dots\dots (2)$$

Step 3 of 8

Consider the transformation ratio.

$$\frac{V_2}{V_1} = \frac{-N_2}{N_1}$$

Substitute 1 for N_1 , and 3 for N_2 .

$$\frac{V_2}{V_1} = \frac{-3}{1}$$

$$V_2 = -3V_1$$

Also,

$$\frac{i_2}{i_1} = \frac{-N_1}{N_2}$$

Substitute 1 for N_1 , and 3 for N_2 .

$$\frac{i_2}{i_1} = \frac{-1}{3}$$

$$i_1 = -3i_2$$

Step 4 of 8

Recall equation (1).

$$\frac{V_1 - 12}{2} + \frac{V_1 - V_2}{\left(\frac{20}{j\omega}\right)} + i_1 = 0$$

Substitute $-3V_1$ for V_2 , 2 for ω , and $-3i_2$ for i_1 .

$$\frac{V_1 - 12}{2} + \frac{V_1 - (-3V_1)}{\left(\frac{20}{j2}\right)} + (-3i_2) = 0$$

$$\frac{V_1 - 12}{2} + j\left(\frac{V_1 + 3V_1}{10}\right) - 3i_2 = 0$$

$$\frac{V_1}{2} - 6 + j\left(\frac{2V_1}{5}\right) - 3i_2 = 0$$

$$V_1\left(\frac{1}{2} + \frac{j2}{5}\right) - 3i_2 = 6 \dots\dots (3)$$

Step 5 of 8

Recall equation (2).

$$-i_2 + \frac{V_2 - V_1}{\left(\frac{20}{j\omega}\right)} + \frac{V_2}{6} = 0$$

Substitute $-3V_1$ for V_2 , and 2 for ω .

$$-i_2 + \frac{-3V_1 - V_1}{\left(\frac{20}{j2}\right)} + \frac{-3V_1}{6} = 0$$

$$-i_2 - j\left(\frac{4V_1}{10}\right) - \frac{V_1}{2} = 0$$

$$V_1\left(\frac{-1}{2} - \frac{j2}{5}\right) - i_2 = 0 \dots\dots (4)$$

Recall equation (4).

$$V_1 \left(\frac{-1}{2} - \frac{j2}{5} \right) - i_2 = 0$$

Multiply with 3.

$$3V_1 \left(\frac{-1}{2} - \frac{j2}{5} \right) - 3i_2 = 0 \dots\dots (5)$$

Recall equation (3).

$$V_1 \left(\frac{1}{2} + \frac{j2}{5} \right) - 3i_2 = 6$$

Subtract equation (3) from equation (5).

$$\left[3V_1 \left(\frac{-1}{2} - \frac{j2}{5} \right) - 3i_2 \right] - \left[V_1 \left(\frac{1}{2} + \frac{j2}{5} \right) - 3i_2 \right] = 0 - 6$$

$$V_1 \left(\frac{-3}{2} - \frac{j6}{5} - \frac{1}{2} - \frac{j2}{5} \right) - 3i_2 + 3i_2 = -6$$

$$V_1 \left(\frac{-4}{2} - \frac{j8}{5} \right) = -6$$

$$V_1 \left(2 + \frac{j8}{5} \right) = 6$$

Simplify further,

$$\begin{aligned} V_1 &= \frac{6}{\left(2 + \frac{j8}{5} \right)} \\ &= \frac{6}{2 + j1.6} \\ &= (1.829 - j1.463) \text{ V} \end{aligned}$$

Determine current i_x .

$$i_x = \frac{V_1 - V_2}{\left(\frac{20}{j\omega} \right)}$$

Substitute $-3V_1$ for V_2 , and 2 for ω .

$$\begin{aligned} i_x &= \frac{V_1 - (-3V_1)}{\left(\frac{20}{j2} \right)} \\ &= \frac{j4V_1}{10} \\ &= \frac{j2V_1}{5} \end{aligned}$$

Substitute $(1.829 - j1.463)$ for V_1 .

$$\begin{aligned} i_x &= \frac{j2(1.829 - j1.463)}{5} \\ &= 0.5852 + j0.7316 \\ &= 0.9368 \angle 51.344^\circ \text{ A} \end{aligned}$$

Convert current i_x into its time-domain.

$$i_x = 0.9368 \cos(2t + 51.344^\circ) \text{ A}$$

Therefore, current i_x is $0.9368 \cos(2t + 51.344^\circ) \text{ A}$.